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AMAZON SALES ANALYSIS

Business Answers & Recommendations

Based on the computational analysis of your **Amazon.csv** dataset, here are the answers and recommendations to the 10 Business Questions from the analysis using Jupyter notebooks and Excel.

1. What is the total sales, total profit, and average order value?

- **Total Sales:** \$91,825,647.92
- **Total Profit:** \$21,059,521.40
- **Average Order Value (AOV):** \$918.26 per transaction

2. Which state generated the highest and the lowest sales?

- **Highest Sales State:** Texas (TX) with \$22,862,540.54 in revenue.
- **Lowest Sales State:** Arizona (AZ) with \$4,504,324.19 in revenue.

3. Which category and sub-category contributed the highest revenue?

- **Highest Category:** Electronics (\$15,584,217.18)
- **Highest Sub-Category/Brand:** CoreTech (\$9,343,720.59)

4. Which are the Top 10 products based on sales?

1. **Memory Card 128GB** (\$1,935,138.40)
2. **LED Desk Lamp** (\$1,921,948.41)
3. **Mechanical Keyboard** (\$1,906,963.54)
4. **Electric Kettle** (\$1,905,751.79)
5. **Smartwatch** (\$1,901,275.59)
6. **Dress Shirt** (\$1,896,765.36)
7. **Water Bottle** (\$1,895,467.95)
8. **Gaming Mouse** (\$1,895,103.98)
9. **Kids Toy Car** (\$1,891,082.43)
10. **Jeans** (\$1,882,591.82)

5. Who are the Top 5 customers contributing the highest revenue?

1. Pooja Kapoor (\$547,832.64)
2. Vihaan Singh (\$526,675.89)
3. Pooja Singh (\$516,451.45)
4. Kabir Joshi (\$515,361.68)
5. Aditya Gupta (\$515,005.09)

6. Which payment mode is most preferred by customers?

- **Credit Card** is the most preferred payment method with **35,038 orders**, followed by Debit Card (20,024 orders). Cash on Delivery is the least used method (4,928 orders).

7. Which month recorded the highest sales and profit? Is there any noticeable sales trend?

- **Highest Sales Month:** May 2024
- **Highest Profit Month:** August 2020
- **Trend Observation:** Sales show dynamic macro-economic cycles rather than simple linear paths. Peaks correspond to mid-year shopping events (May) and holiday inventory clearance windows.

8. Identify any categories or products with high sales but low profit. What could be the possible reason?

- **Analysis:** The **Electronics** category generated the single highest revenue stream (\$15.58M) but its final profit margin (22.92%) is lower than categories like *Toys & Games* (22.95%) or *Clothing* (22.94%).
- **Possible Reasons:** Electronics encounter severe customer acquisition costs, high unit return rates (due to functional issues/buyer remorse), aggressive competitive discounting, and significantly heavier shipping/logistical costs which eat into margins.

9. Which state or city requires improvement in sales performance? Suggest possible business strategies.

- **Target Region:** Arizona (AZ) is performing poorly, bringing in less than a fifth of what Texas brings in.
- **Strategies:** 1. Allocate localized micro-marketing campaigns targeted around key tier-2 cities in Arizona. 2. Onboard and build incentives for local state-level sellers to offer faster standard shipping times, eliminating transit friction.

10. Three Actionable Business Recommendations for Amazon Management

1. Capitalize on Credit Card Preferences via Premium Co-Branding

- **Insight:** With over 35% of total transactions originating from Credit Cards, credit users form your primary engine.
- **Action:** Partner with major banks in India to offer specialized co-branded credit card rewards (e.g., immediate 5% cashback on high-revenue brands like *CoreTech*). This creates a direct incentive to drive up your \$918 Average Order Value.

2. Re-engineer Logistics for Bulky/Complex Categories

- **Insight:** Electronics capture the highest volume but lose efficiency to localized margin degradation.
- **Action:** Establish focused fulfillment pipelines for high-demand items like *Memory Cards* and *Keyboards*. Moving these items to fulfillment hubs closer to key volume states like Texas lowers shipping cost overheads and maximizes net margins.

3. Target Marketing Budgets on Top 5% Customer Archetypes

- **Insight:** Top individual shoppers like *Pooja Kapoor* spend over half a million rupees individually, showing massive customer lifetime value (LTV).
- **Action:** Pivot marketing operations away from cold audience generation toward customer-retention strategies. Introduce exclusive high-tier loyalty status clubs providing early product access, zero shipping costs, and targeted product recommendations to your top historical spenders.